

Industrial Internship Report on "Password Manager"

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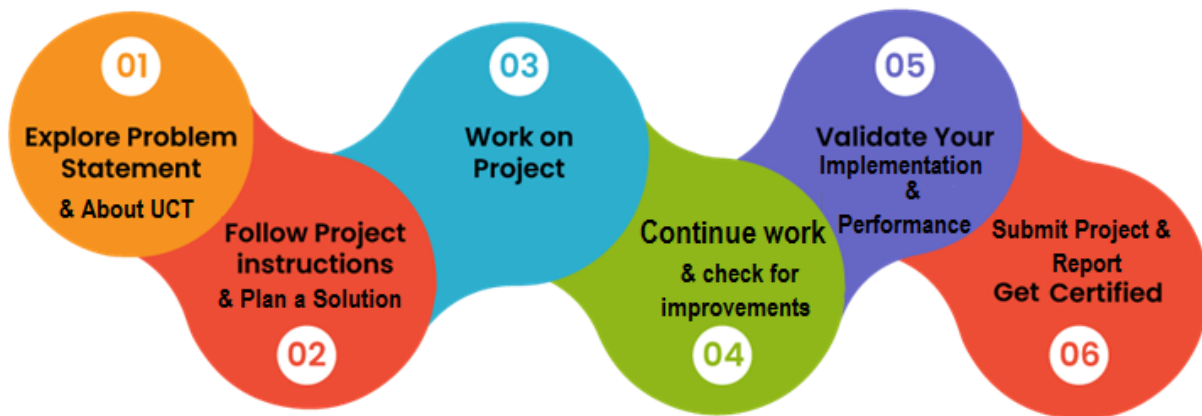
Executive Summary

My project was **Password Manager using Python**. The objective of this project was to develop a secure and user-friendly application for storing, managing, and retrieving passwords efficiently. Throughout the internship, I learned Python programming fundamentals, file handling, conditional statements, functions, NumPy, and Pandas. These technologies helped me understand data processing, program organization, and efficient coding practices. This internship provided valuable practical exposure to software development and strengthened my problem-solving and analytical skills while working on a real-world application.

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1 Preface



This internship was conducted over six weeks under the guidance of Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. The internship focused on developing a Password Manager application using Python while learning important Data Science concepts.

During the internship, I gradually learned Python programming, conditional statements, loops, functions, file handling, NumPy, and Pandas. Each week introduced new concepts which were applied to improve the project.

This internship enhanced my technical knowledge, logical thinking, debugging skills, and software development practices. I sincerely thank Upskill Campus, UniConverge Technologies Pvt. Ltd., my mentors, and faculty members for their valuable guidance and support throughout this internship.

I encourage future students to make full use of such internship opportunities because they provide practical knowledge and improve confidence in solving real-world problems.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



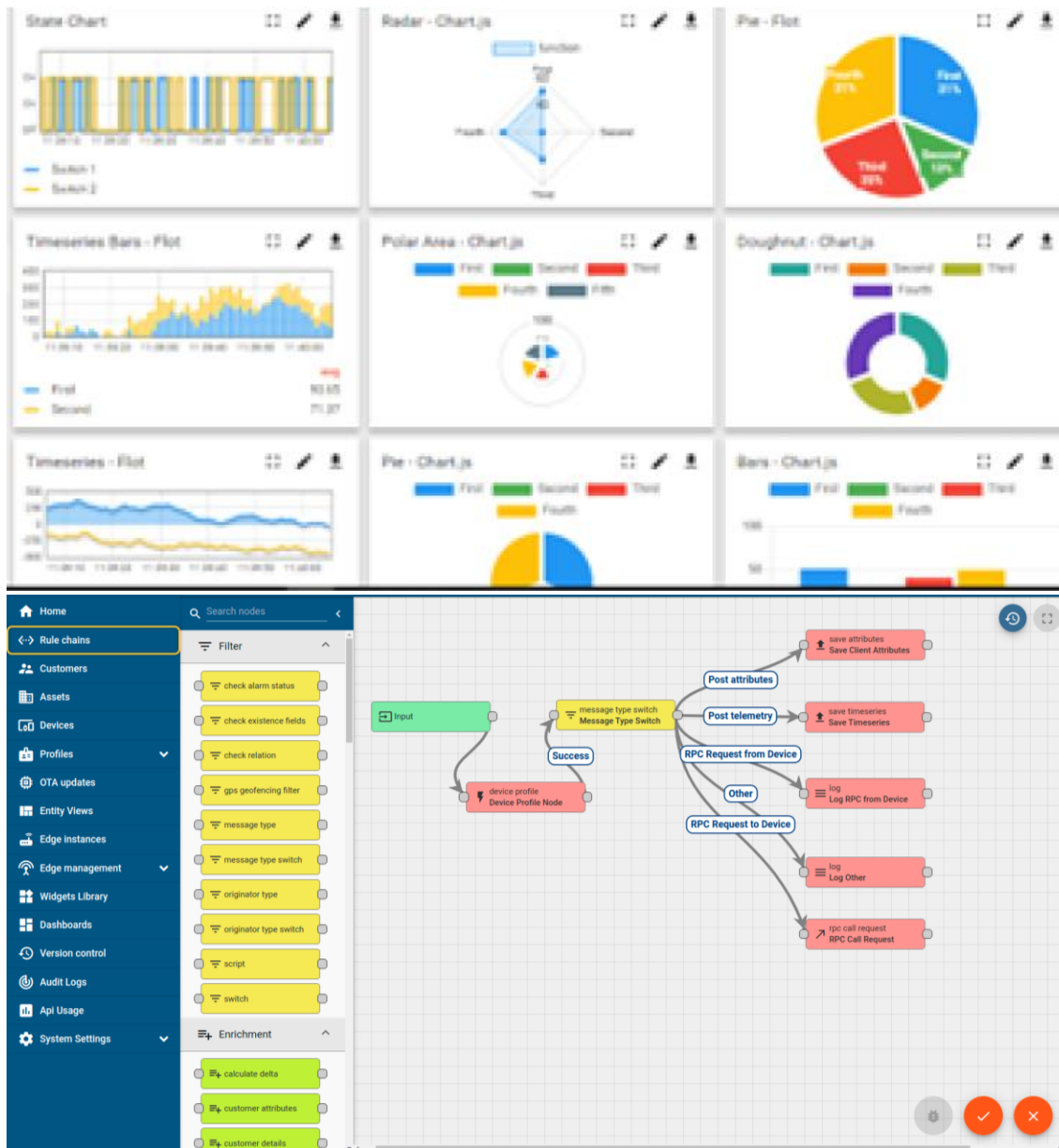
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (**FACTORY WATCH**)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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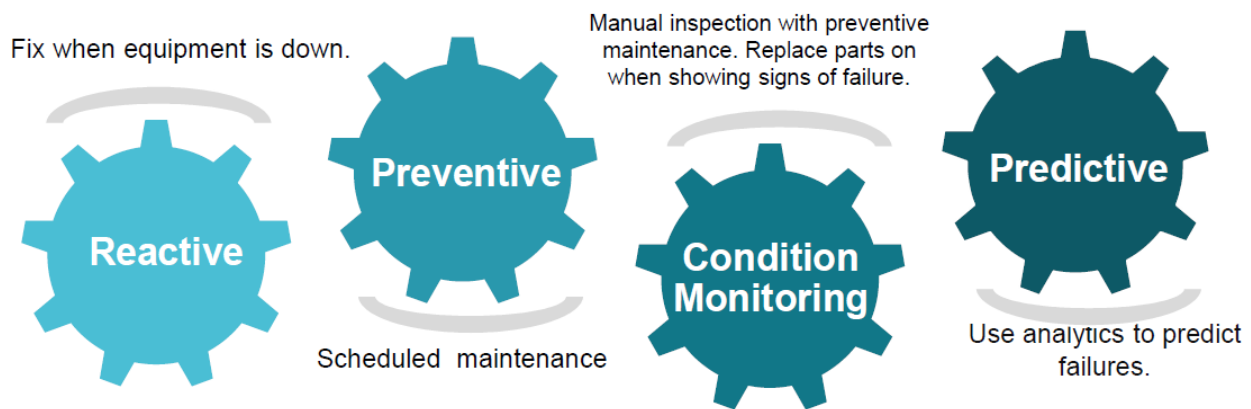


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

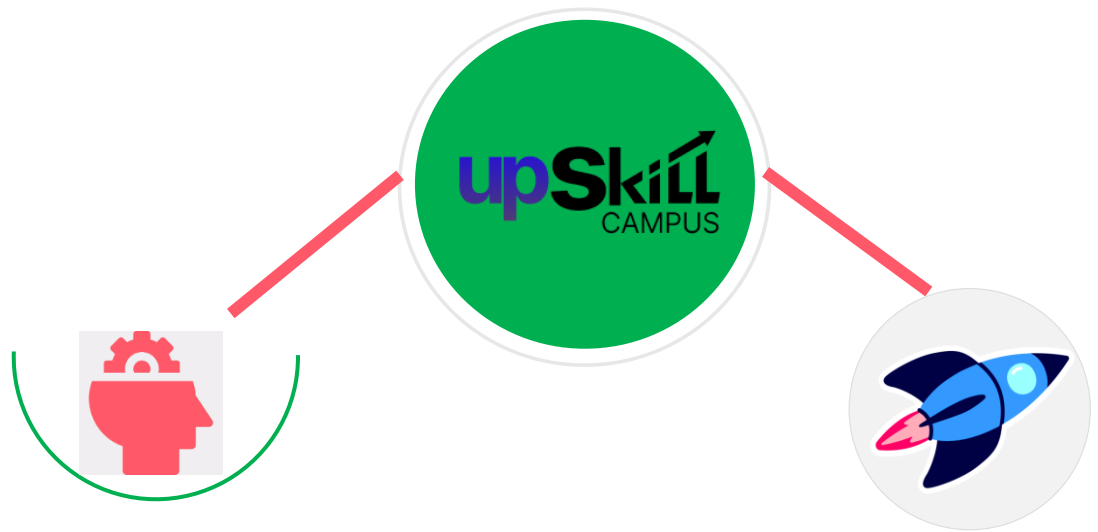
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

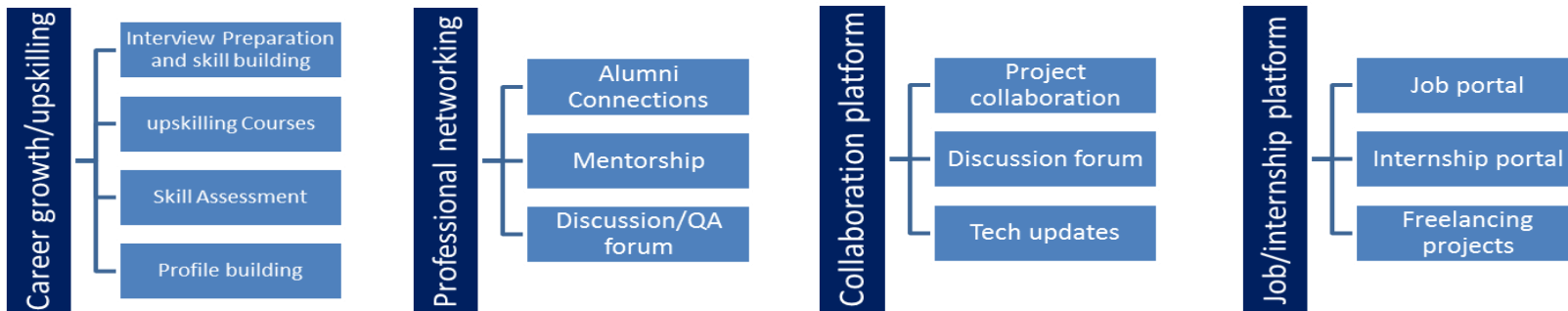
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] <https://docs.python.org/3/>

[2] <https://numpy.org/doc/>

[3] <https://pandas.pydata.org/docs/>

2.6 Glossary

Python -

NumPy Numerical Python

Pandas Python Data Analysis Library

Terms	Acronym

3 Problem Statement

Managing multiple online accounts has become increasingly difficult due to the large number of usernames and passwords users need to remember. Many users reuse weak passwords or store them insecurely, increasing security risks. The objective of this project is to develop a Password Manager using Python that allows users to securely store, retrieve, update, and manage passwords through a simple menu-driven application. The project also introduces efficient data handling techniques using Python, NumPy, and Pandas to improve data organization and processing.

4 Existing and Proposed solution

Most users store passwords in notebooks, text files, or browser password managers. These methods either lack proper security, are difficult to manage, or are platform-dependent.

The proposed Password Manager provides a lightweight Python-based solution that enables users to securely store, view, update, search, and delete passwords. The application uses file handling for persistent storage and applies Python programming concepts to create a simple and efficient system.

4.1 Code submission (Github link)

<https://github.com/teenaa77/upskill-campus/tree/main>

4.2 Report submission (Github link) : [https://github.com/teenaa77/upskill-campus/blob/main/Password %20Manager%20 USC UCT.pdf](https://github.com/teenaa77/upskill-campus/blob/main/Password%20Manager%20USC%20UCT.pdf)

5 Proposed Design/ Model

User

|



Password Manager

|

|— Add Password

|— View Password

|— Search Password

|— Update Password

|— Delete Password

|



Password Storage File

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

Start



Display Menu



User Choice



Perform Operation



Save Changes



Exit

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

The Password Manager application was tested to evaluate its functionality, reliability, and correctness under different user scenarios. The objective was to ensure that all major features worked accurately and that the program handled user inputs effectively.

6.1 Test Plan/ Test Cases

Tested all menu options using different user inputs

6.2 Test Procedure

Executed the application with valid and invalid inputs to verify functionality.

6.3 Performance Outcome

The application successfully performed password storage, retrieval, update, and deletion operations with accurate results. The program handled invalid inputs correctly and maintained consistent data storage.

7 My learnings

. This internship significantly improved my understanding of Python programming and Data Science fundamentals. I learned file handling, conditional statements, loops, functions, NumPy, and Pandas through practical implementation. Working on the Password Manager project enhanced my logical thinking, debugging skills, and software development knowledge. The internship also strengthened my confidence in solving real-world programming problems and prepared me for future technical projects and career opportunities.

Future work scope

The Password Manager can be further enhanced by adding advanced security features such as password encryption and user authentication. Future improvements may also include a graphical user interface (GUI), password strength analysis, automatic password generation, cloud database integration, and multi-user support. These enhancements would improve the security, usability, and scalability of the application, making it more suitable for real-world use.